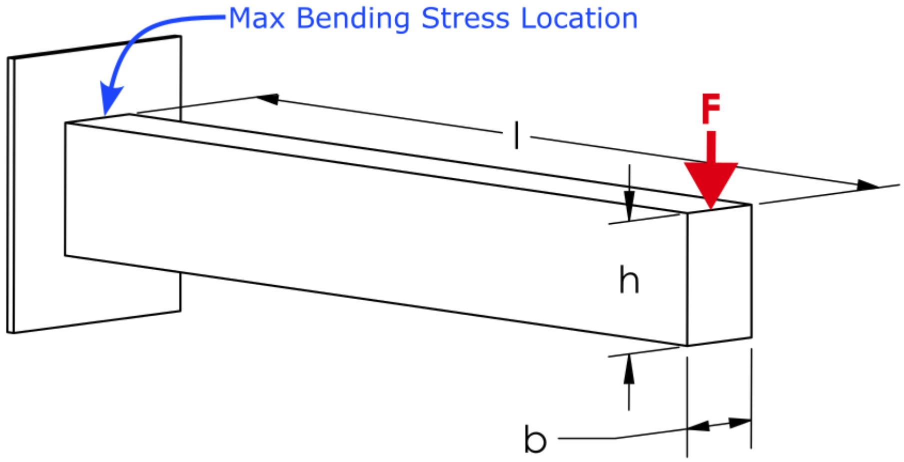
Database Consistency Test

Introduction

This EngineeringPaper.xyz sheet shows how to calculated the maximum stress and displacement for a cantilever beam loaded by a vertical force at its end. The geometry of the beam is shown in the figure below. The beam has a rectangular cross section with height h and width b. The maximum stress in a cantilever with length much larger than its height is due to bending stress that occurs at the base of the beam. For a downward force, the maximum stress will be a tensile force at the top of the beam and a compressive stress at the bottom of the beam. For beam sections that are symmetric top to bottom, the maximum tensile and compressive stresses will have equal magnitude. μ ▯ #



Select Beam Material

$$\text{Aluminum Alloys} \begin{cases} E &= 71.7\,[GPa] \\ v &= 0.34 \\ \rho &= 2800\left[\frac{kg}{m^3}\right] \end{cases}$$

Source: Norton, R. L. "Machine design. A integrated approach, 5th Editi." (2013).

Select Beam Section

$$\text{Rectangle}\quad\begin{cases} I_x & = & \frac{b \cdot h^3}{12} \\ c & = & \frac{h}{2} \\ r & = & [mm] \\ r_1 & = & [mm] \\ r_2 & = & [mm] \\ a & = & [mm] \\ b & = & 50\,[mm] \\ h & = & 75\,[mm] \\ b_1 & = & [mm] \\ h_1 & = & [mm] \end{cases}$$

Source for beam section equations and images: https://en.wikipedia.org/wiki/List_of_second_moments_of_area

Define the Other Beam Parameters

$$l = 500\,[mm]$$

$$F = 500\,[N]$$

Define the Deflection and Stress Equations

The maximum displacement of a cantilever beam loaded at its end is defined by [1]:

$$y_{max} = -\frac{F \cdot l^3}{3 \cdot E \cdot I_x}$$

where Ix is the area moment of inertia of the cross section about the x-axis as selected in the table above. The maximum displacement can now be queried:

$$y_{max} = [mm] = -0.165297794307557\,[mm]$$

The maximum bending stress in a beam section is given by:

$$\sigma_{max} = \frac{M \cdot c}{I_x}$$

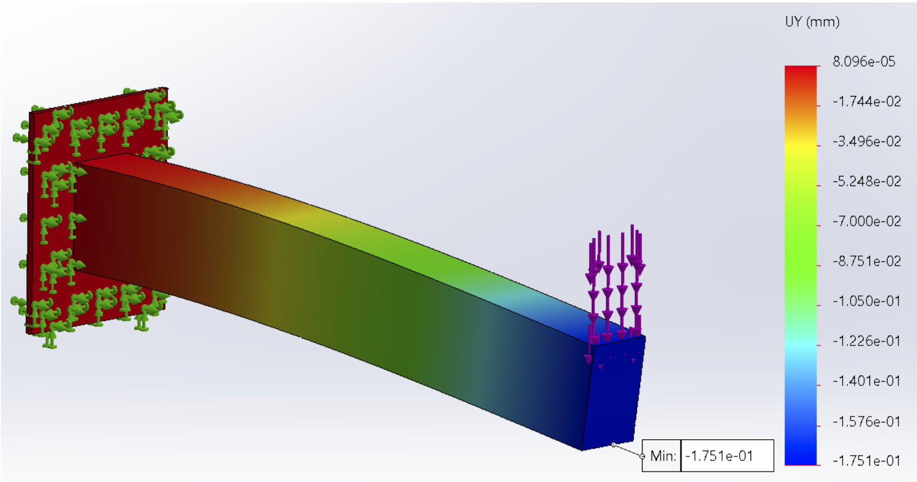
where M is the bending moment at the section of interest and c is distance from the neutral axis of the beam to the top or bottom of the beam. The maximum moment occurs at the base of the beam and is defined by:

$$M = l \cdot F$$

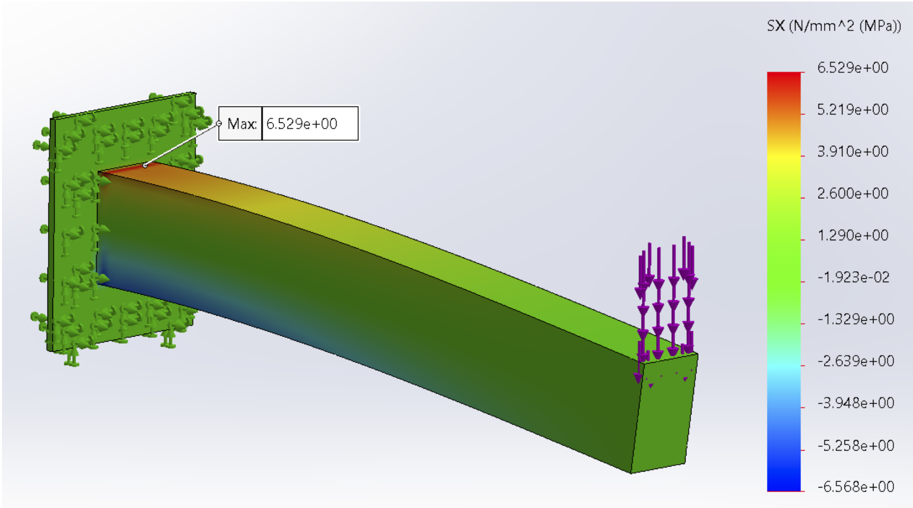
where l is the length of the beam. Now that everything is defined, the maximum stress can be queried and converted to [MPa] units:

$$\sigma_{max} = [MPa] = 5.33333333333333\,[MPa]$$

These results can be compared to the results obtained from a finite element analysis using SolidWorks. The beam has the same dimensions, loading, and material as the rectangular cross-section and the aluminum table options above. The displacement plot is shown below. The maximum displacement magnitude obtained using finite element analysis is .175 [mm], which is close to the .165 [mm] obtained using the equations above.

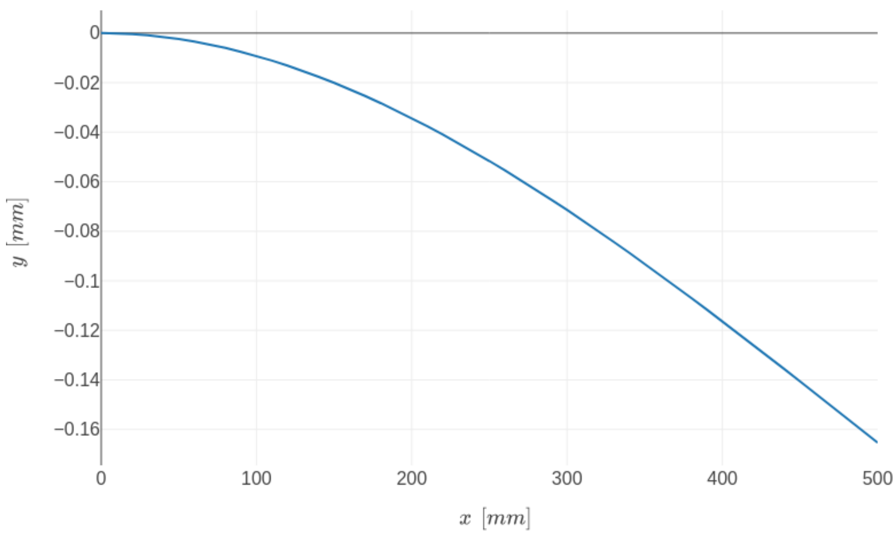


Similarly, the maximum stress calculation can be compared to the SolidWorks results. The maximum stress obtained from SolidWorks is 6.5 [MPa] as shown below. Note that this is higher than the 5.3 [MPa] obtained using the beam equation. This is the result of the stress concentration that occurs at the based of the cantilever where it meets the fixed boundary condition. A littler further from the fixed boundary condition, the finite element values are close to the calculated values.



[1] Norton, R. L. "Machine design. A integrated approach, 5th Editi." (2013).

$$y = \frac{F}{6 \cdot E \cdot I_x} \cdot \left(x^3 - 3 \cdot l \cdot x^2\right)$$



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